

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal Sentinel-1 SLC IW mode data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 1006+950

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

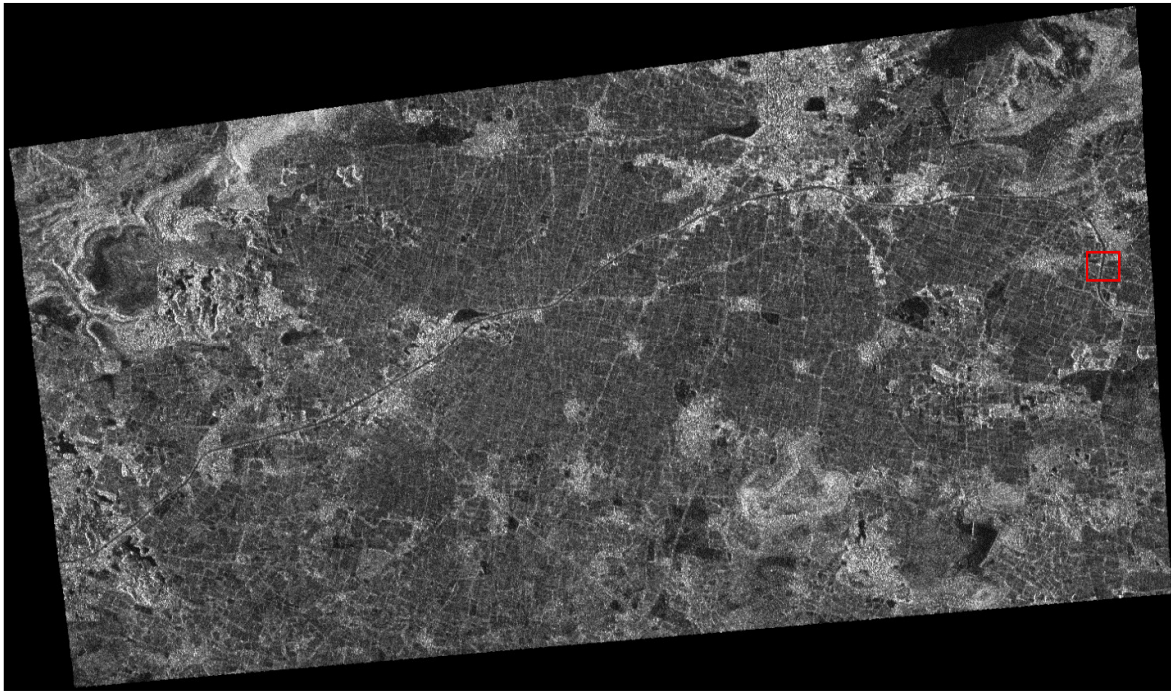
Flight Direction - DESCENDING

Polarization – VV+VH

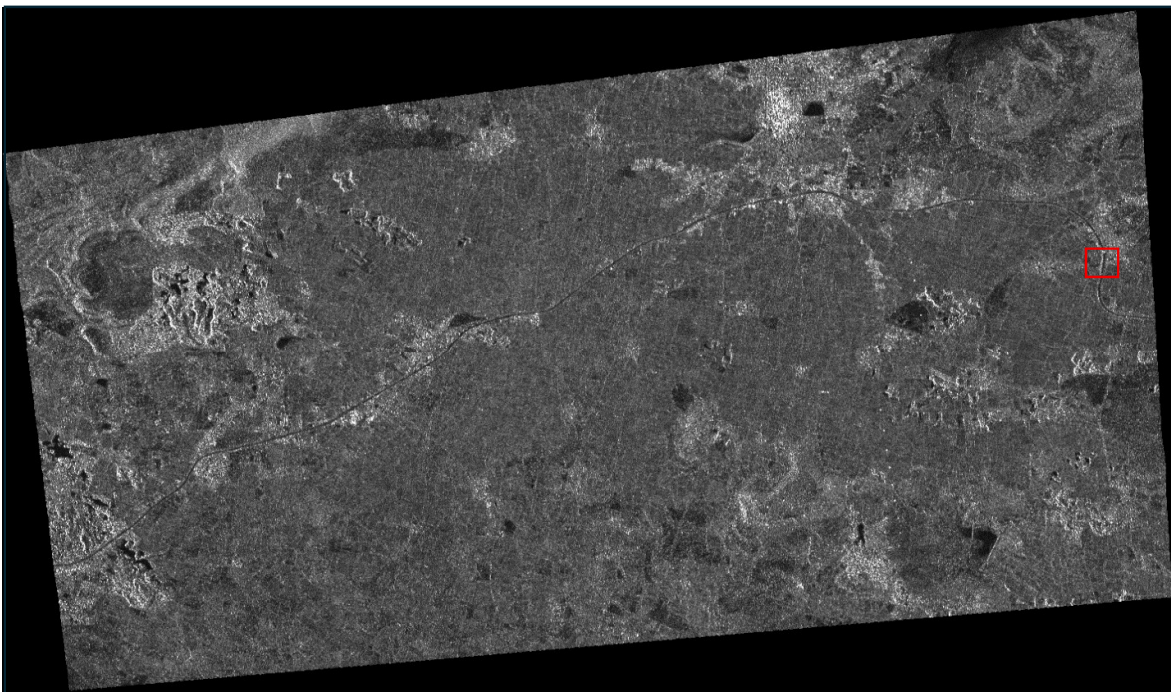
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

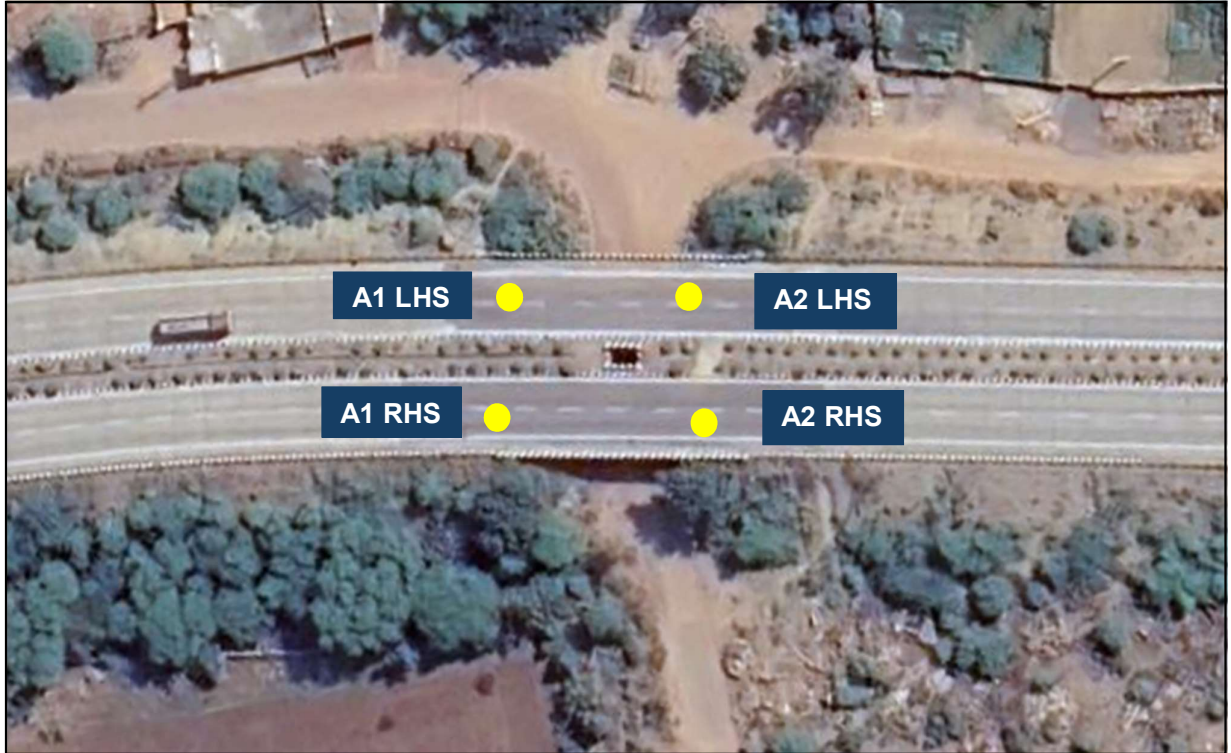


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

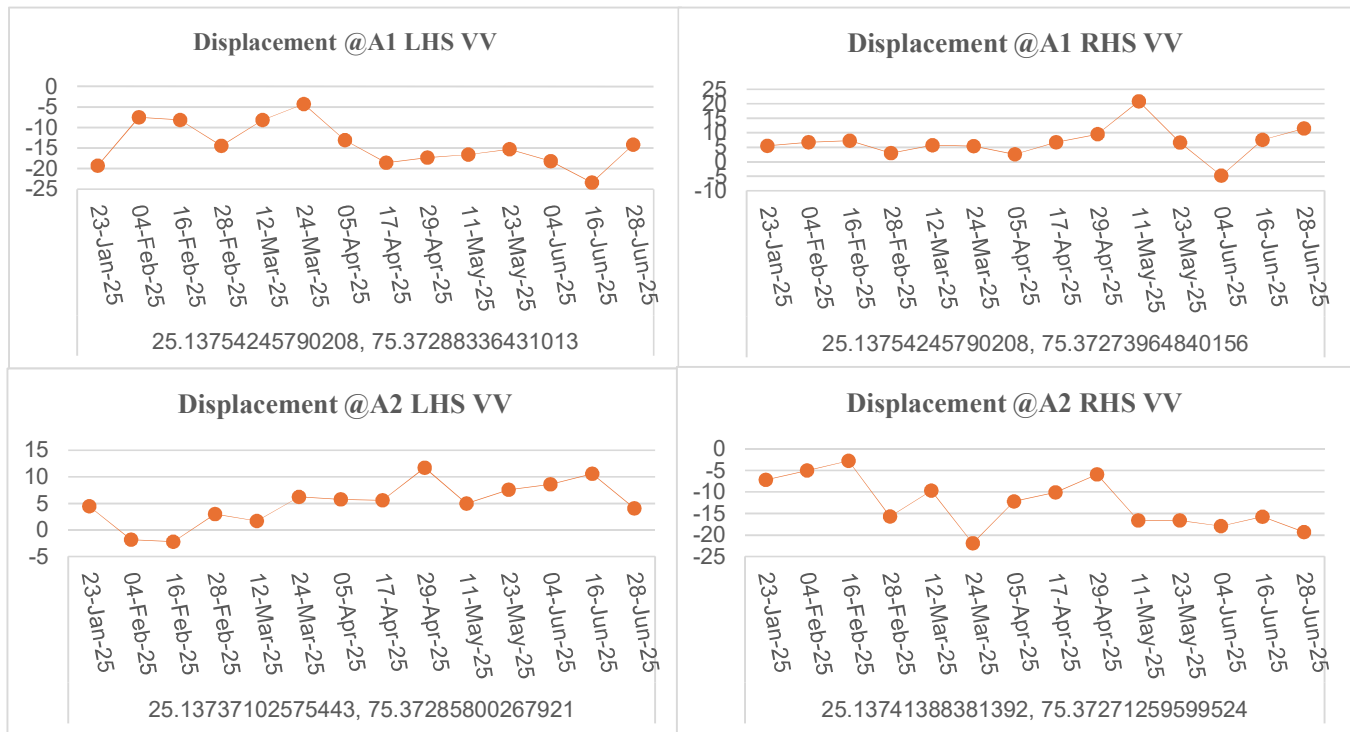
VUP – CH-1006+950

Length – 22.34M

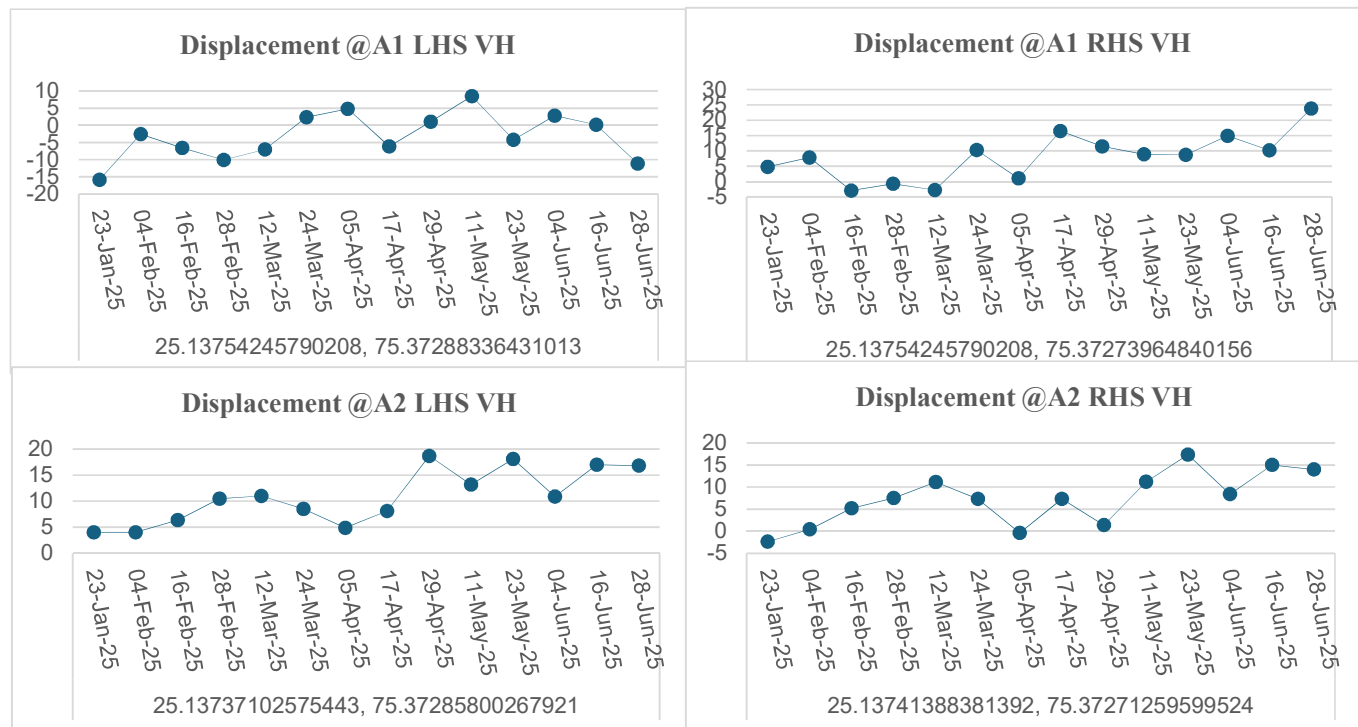


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



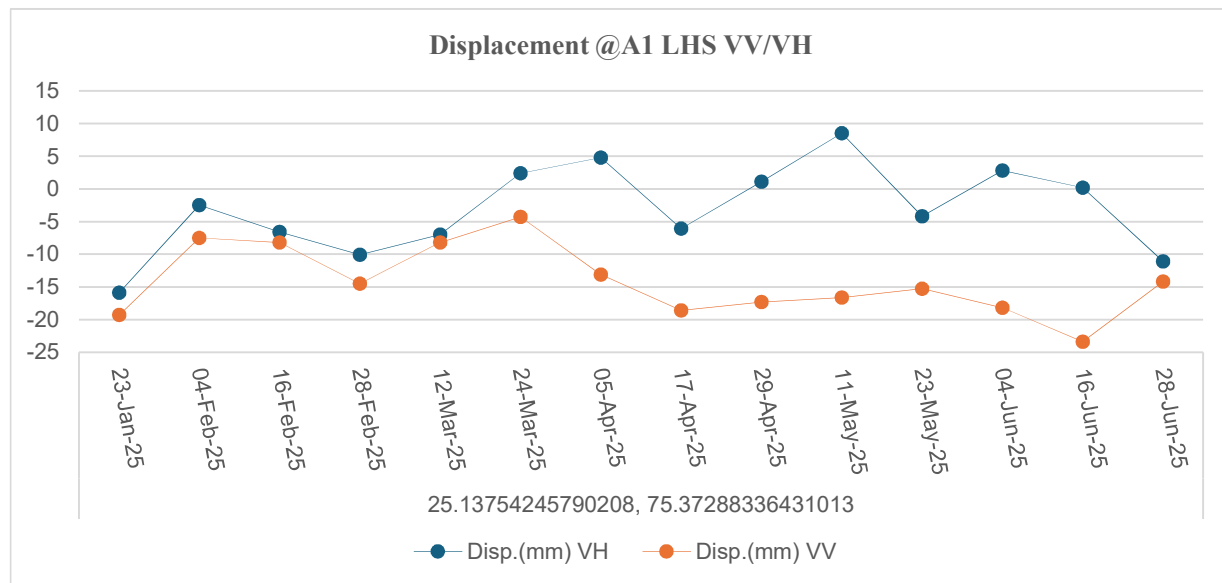
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.13754245790208, 75.37288336431013)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** remains predominantly negative throughout the monitoring period, varying from about -7 mm to -24 mm.
- **Horizontal Displacement (VH)** at A1 LHS shows clear oscillatory behaviour, ranging approximately from -15 mm to +8 mm.

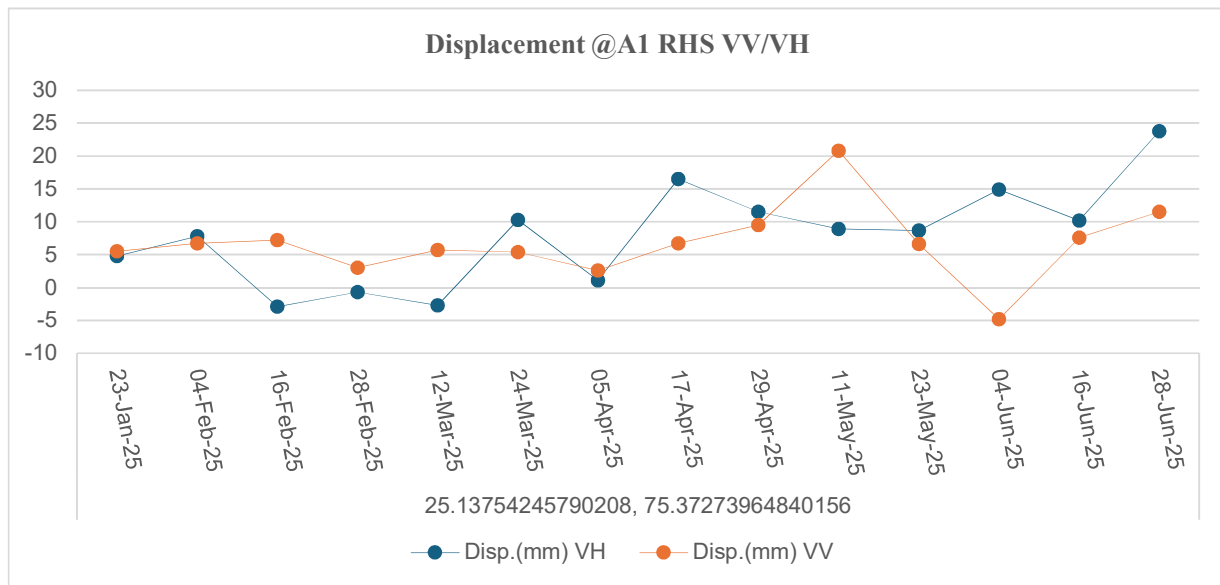
Critical Events & Anomalies:

- **January 2025:** VH begins at around -15 mm, while VV is approximately -18 mm, indicating initial lateral displacement accompanied by significant vertical settlement, suggesting early-stage support or foundation response.
- **June 2025:** Both VH and VV decline again, with VH reaching -11 mm and VV approaching -24 mm, indicating continued settlement and renewed lateral movement, pointing to progressive deformation rather than recovery.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.13754245790208, 75.37273964840156)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** at A1 RHS shows moderate to high variability, ranging approximately from –4 mm to +24 mm.
- **Vertical displacement (VV)** varies between approximately –5 mm and +21 mm, showing alternating settlement and uplift behaviour.

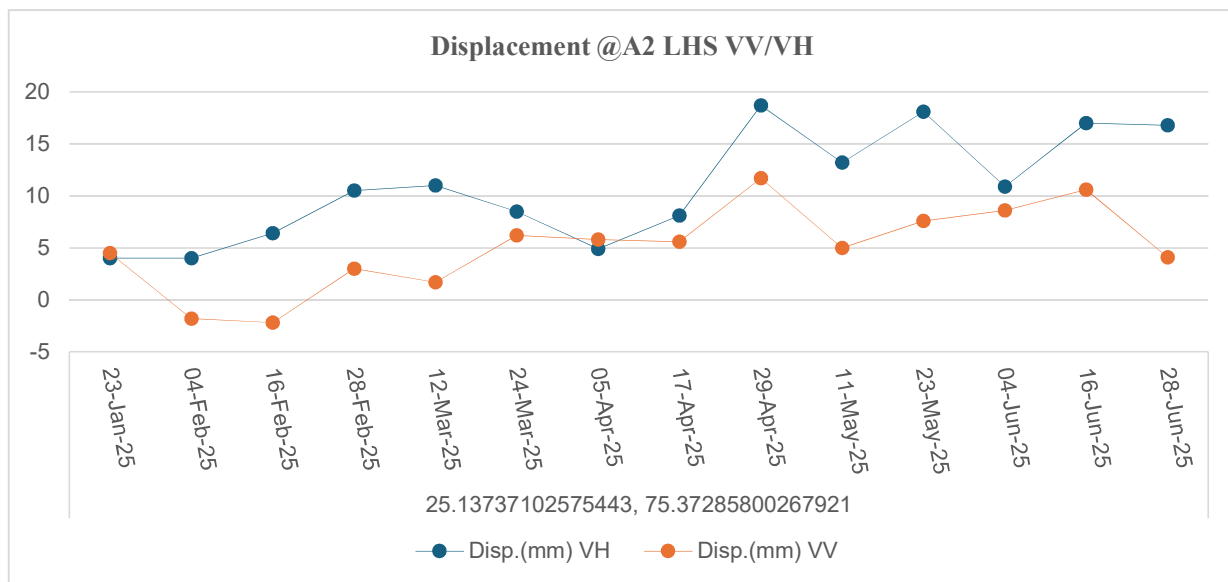
Critical Events & Anomalies:

- **January 2025:** VH and VV both start near +5 mm to +6 mm, indicating initially stable lateral and vertical conditions with minor uplift.
- **June 2025:** VH reaches its maximum of approximately +24 mm, and VV recovers to around +12 mm, indicating continued lateral progression with partial vertical recovery.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.13737102575443, 75.37285800267921)



Key Observations:

Displacement Magnitude Patterns:

- Horizontal displacement (VH)** at A2 LHS shows a clear increasing trend over time, ranging approximately from +4 mm to +19 mm.
- Vertical Displacement (VV)** varies between approximately –2 mm and +11 mm, showing alternating settlement and uplift behaviour.

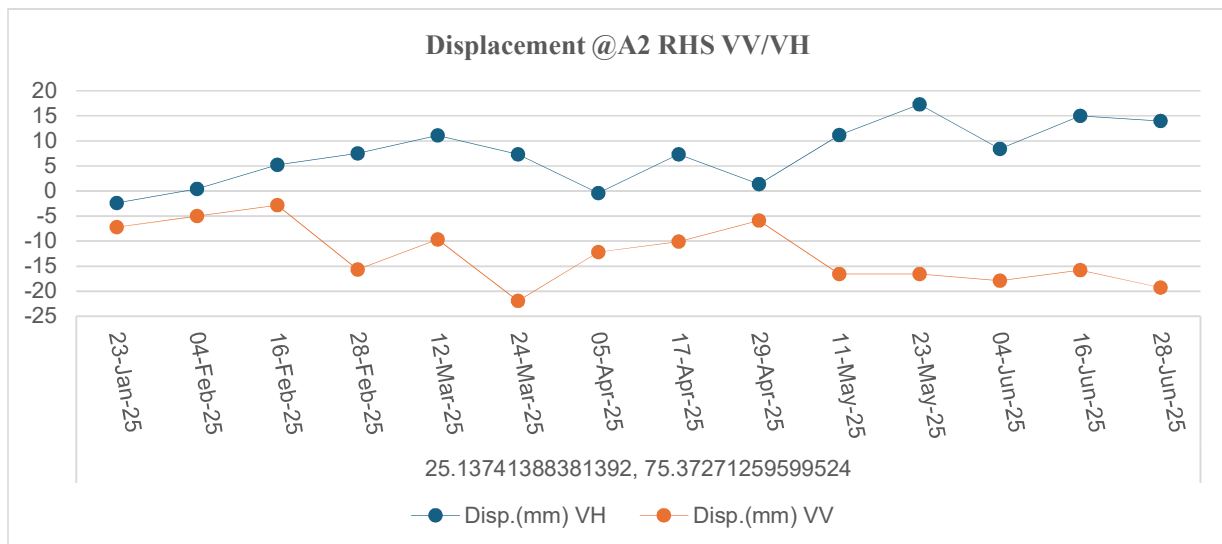
Critical Events & Anomalies:

- January 2025:** Both VH and VV begin around +4 mm, indicating initially stable lateral and vertical conditions at A2 LHS.
- June 2025:** VH stabilizes at a high level (16–17 mm), while VV peaks again near +10 mm before dropping to +4 mm by late June, indicating continued lateral dominance with vertical oscillation.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.13741388381392, 75.37271259599524)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement is predominantly negative, ranging roughly between -5 mm and -22 mm, indicating continuous settlement at A2 RHS.
- **Horizontal (VH) displacement** at A2 RHS shows a clear progressive increasing trend, ranging approximately from -2 mm to +18 mm.

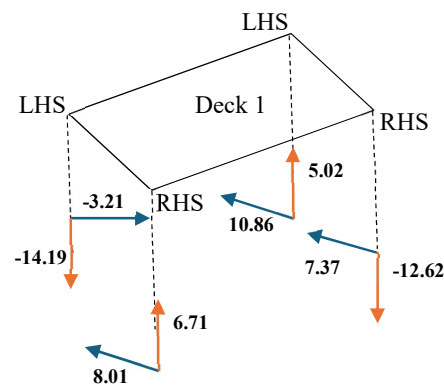
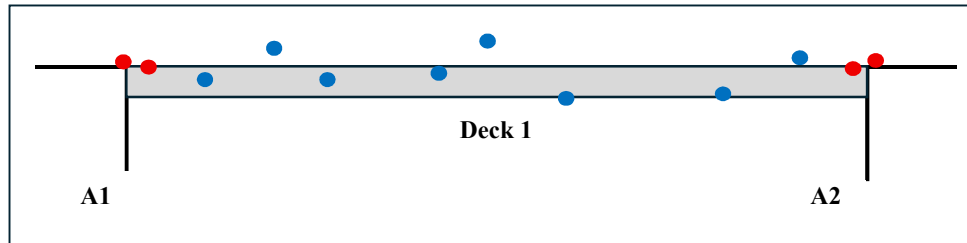
Critical Events & Anomalies:

- **January 2025:** VH starts near -2 mm, while VV is around -6 mm, suggesting initial minor lateral displacement accompanied by vertical settlement.
- **April 2025:** VH increases to approximately +11 mm, while VV recovers slightly to around -5 mm, suggesting partial vertical rebound with continued horizontal drift.
- **June 2025:** VH remains elevated (14-15 mm), while VV continues to decline toward -20 mm, confirming persistent settlement dominance in the vertical direction.

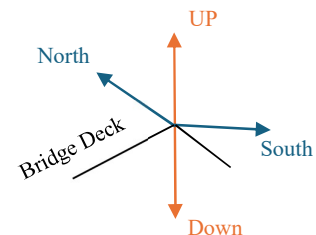
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits highly non-uniform vertical displacement, indicating strong differential vertical response between the LHS and RHS regions. The coexistence of uplift at the deck core and settlement at both edges confirms vertical rotation combined with localized bending, rather than uniform sagging.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates pronounced directional variability, clearly reflecting torsional deformation of the bridge deck slab. The presence of opposing directions and uneven magnitudes across LHS and RHS confirms torsional rotation about a longitudinal axis, rather than rigid-body translation.

CONCLUSION:

- The bridge deck slab demonstrates a torsion-dominated deformation response, characterized by severe vertical settlement at both supports combined with uplift at the deck core. Overall, the magnitude and asymmetry of the observed displacements indicate structurally significant behaviour, warranting continued monitoring and detailed assessment, particularly with respect to torsional effects and bearing performance.